Coastal Human Surveillance: Dual-Zone Pedestrian and Swimmer Risk Analytics via High-Resolution Vision Detectors

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***Abstract*—Automated computer vision surveillance in dynamic marine environments represents a critical yet underserved frontier in maritime public safety. Rip currents, sudden tidal surges, and unpredictable surf conditions account for over 100,000 ocean rescues worldwide on an annual basis. Conventional lifeguard surveillance remains constrained by human visual fatigue, environmental solar glare, and restricted physical line-of-sight from static elevated observation towers. Standard automated video surveillance systems fail in this domain due to severe scale variance between tiny distant swimmer heads ($< 16 \times 16$ pixels) and large foreground beachgoers (> 120 pixels), relentless marine clutter from breaking surf, and an utter absence of dual-zone spatial intelligence separating safe beachgoers on sand from endangered bathers in deep water. This paper establishes a rigorous, multi-paradigm deep learning benchmark for automated coastal human detection and dual-zone spatial analytics. Evaluating on a curated, noise-filtered corpus of 1131 high-resolution coastal scenes comprising over 15,900 verified human bounding boxes, we systematically investigate five state-of-the-art vision architectures: RT-DETR (Real-Time Vision Transformer with Hybrid Encoder), YOLOv8x-P2 (High-Resolution Stride-4 Feature Head), RetinaNet (Single-Stage Focal Loss ResNet-50), Faster R-CNN (Two-Stage ResNet-50 with RoIAlign), and Detectron2 (Faster R-CNN with Deep ResNet-101 FPN). Across standardized benchmarks on an unseen test corpus, RT-DETR achieves the highest overall accuracy with 94.86% mAP@0.50, 84.25% strict mAP@0.50:0.95, and 91.51% Mean IoU. YOLOv8x-P2 demonstrates superior edge viability, achieving 46.2 FPS on an NVIDIA Tesla T4 GPU alongside a 98.12% global headcount accuracy. Detectron2 ResNet-101 defines the benchmark localization ceiling at 94.15% mAP@0.50, sustaining a 97.7% Land purity and 95.0% Water purity in target confusion matrix evaluations. Furthermore, we contribute two formal theoretical formulations: a continuous Dynamic Shoreline Spatial Hazard Metric ($H_i(t)$) and a Scale-Adaptive Multi-Pyramid Loss function (L_{Coastal}) that systematically bridge perceptual bounding box detections with actionable, automated marine hazard warning systems.**

***Index Terms*—Coastal surveillance, dual-zone classification, RT-DETR, YOLOv8-P2, Detectron2, Faster R-CNN, RetinaNet, maritime safety, deep learning, shoreline hazard analytics.**

This research was conducted as part of the Senior Capstone Research Initiative in the Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Coimbatore, India.

I. Introduction

Global coastal and marine recreation represents an indispensable socio-economic activity, yet it is accompanied by acute environmental hazards. Dangerous rip currents, unexpected drop-offs, and dynamic surf swells are responsible for tens of thousands of critical drowning emergencies each year. The operational standard for coastal preservation continues to rely on manual human lifeguards stationed in elevated watchtowers or conducting shoreline foot patrols. However, human visual attention deteriorates sharply over extended observation periods. Severe environmental factors such as blinding solar glint reflecting off water ripples, thermal atmospheric haze, heavy marine fog, and line-of-sight occlusions caused by dense beach umbrellas severely impede human visual reliability.

While the deployment of Unmanned Aerial Vehicles (UAVs) and fixed high-resolution Closed-Circuit Television (CCTV) cameras has expanded across beach coastlines, automated video analysis remains exceptionally brittle when deploying conventional off-the-shelf object detectors. The coastal surveillance domain presents four fundamental computer vision challenges that confound standard models:

1) **Extreme Object Scale Disparity:** Bathers walking on dry foreground sand appear as full-body, high-resolution silhouettes exceeding 120 pixels in height. Conversely, swimmers and surfers venturing into deeper waters present only exposed heads or partial upper torsos, frequently occupying fewer than 16×16 pixels against a turbulent ocean backdrop. Standard feature downsampling schemes obliterate these tiny spatial representations.
2) **Aggressive Marine and Environmental Clutter:** Cresting surf, turbulent white-water foam trails, sea spray, and shimmering solar reflections continuously generate high-frequency textural patterns that trigger

widespread false positive alarms in standard feature extraction backbones.

3) **Severe Crowd Density and Overlapping Occlusions:** On recreational shores, bathers frequently cluster in dense groups along the water margin or beneath sunshades. Traditional Non-Maximum Suppression (NMS) algorithms routinely merge distinct adjacent individuals into single inaccurate bounding boxes.
4) **Absence of Dual-Zone Spatial Semantics:** Classical pedestrian detection systems treat every detected individual uniformly as an generic "person" class. In marine safety operations, an individual relaxing on dry sand is in a state of absolute safety, whereas an individual submerged in ocean surf is exposed to rip-current dynamics. Without automated semantic differentiation between dry beach land and active aquatic zones, intelligent emergency alerting is impossible.

To address these challenges, this paper develops an automated end-to-end framework for dual-zone pedestrian and swimmer risk analytics. The primary technical and empirical contributions of this work are summarized as follows:

- **Rigorous Coastal Dataset Curation and Cleansing:** We assemble and cleanse a specialized high-resolution dataset of 1131 marine scenes containing 15,900+ labeled bounding box instances. Through systematic forensic auditing, we eliminate 1,874 non-human confounding annotations (e.g., traffic cones, beach umbrellas, sun loungers, and marine buoys), enforcing pure dual-class categorization into *Human (Land)* (Class 0) and *Coastal (Water)* (Class 1).
- **Comprehensive Multi-Paradigm Deep Learning Benchmark:** We systematically implement, train to full convergence, and evaluate five diverse state-of-the-art vision architectures under identical, standardized input resolutions (960×960) and hardware parameters: RT-DETR (Transformer), YOLOv8x-P2 (Stride-4 Anchor-Free), RetinaNet (Focal Loss ResNet-50), Faster R-CNN (Two-Stage ResNet-50), and Detectron2 (Deep ResNet-101 FPN).
- **Novel Theoretical Formulations for Marine Risk:** We mathematically formalize two project-specific theoretical contributions: (i) a continuous *Dynamic Shoreline Spatial Hazard Metric* ($H_i(t)$) that computes real-time bather danger as a function of shoreline proximity and localized crowd surge density, and (ii) a *Scale-Adaptive Multi-Pyramid Loss* (L_{Coastal}) that applies an inverse scale amplifier to prevent micro-swimmer gradient attenuation.
- **Extensive Empirical Validation and Trade-Off Analysis:** We conduct exhaustive quantitative evaluations reporting COCO metrics (AP_{50}, AP_{75}, AP_{small}, AP_{medium}, AP_{large}, AR), class-wise confusion matrices, inference frame rates, and visual ground-truth verifications, delineating the operational Pareto frontier for real-time edge UAVs versus forensic ground-station servers.

II. Related Work

A. Real-Time Vision Transformers and Set Prediction

Object detection has experienced a paradigm shift from hand-crafted anchor clustering toward end-to-end transformer-based formulations. Carion et al. pioneered DEtection TRansformer (DETR), treating detection as a direct set prediction problem optimized via Hungarian bipartite matching. However, original DETR models suffered from slow training convergence and extreme computational complexity. Addressing this, Zhao et al. [1] engineered RT-DETR, introducing an Efficient Hybrid Encoder that decouples intra-scale self-attention from cross-scale feature fusion. By eliminating Non-Maximum Suppression (NMS), RT-DETR eliminates clustering latencies and false box merges in dense crowds, achieving 53.1% mAP on COCO. Lv et al. [2] further proposed RTS-DETR for small target detection, employing multi-scale cross-attention with high-resolution queries, demonstrating that attention mechanisms can capture distant human targets within complex outdoor crowds.

B. High-Resolution Stride-4 Heads in Aerial Surveillance

Single-stage YOLO architectures have dominated real-time embedded surveillance due to their aggressive downsampling backbones and anchor-free regression heads. However, standard YOLO models typically operate with a maximum feature map resolution of Stride-8 (P_3, 114×114 at input 640×640). Wang et al. [3] and subsequent UAV surveillance literature demonstrated that Stride-8 features suffer from severe spatial information loss when applied to tiny targets under 16 pixels. By introducing an additional high-resolution Stride-4 (P_2, 229×229) feature pyramid level into YOLOv8, spatial gradients from distant micro-targets are preserved through the feature aggregation neck, directly enhancing recall on distant persons by over 7.3% [4].

C. Marine Background Clutter and Dynamic Loss Re-Weighting

Marine computer vision involves extreme foreground-to-background spatial class imbalance. In a typical high-altitude coastal video frame, dry sand expanses and open water waves comprise more than 99% of all pixels, while human targets occupy a minuscule spatial fraction. Lin et al. [11] formulated Focal Loss, introducing a dynamic modulating factor $(1-p_t)^\gamma$ to downweight the loss contribution of easily classified background negative tiles. Zhang et al. [5] and Liu et al. [6] adapted dynamic focal formulations to maritime Synthetic Aperture Radar (SAR) and optical target detection, demonstrating that setting focusing parameter $\gamma = 2.0$ reduces sea-spray and white-cap wave false alarm rates by up to 74%.

D. Two-Stage Proposal Architectures and Deep Feature Hierarchies

Two-stage frameworks remain the gold standard for localization precision. Ren et al. [12] introduced Faster R-CNN, decoupling detection into class-agnostic candidate region proposals generated by a Region Proposal Network

(RPN), followed by RoI pooling and classification. Chen et al. [7] established that pairing Faster R-CNN with RoIAlign eliminates quantization artifacts, preserving exact sub-pixel spatial coordinates for partially submerged targets. Wu et al. [9] formalized the Detectron2 modular framework, demonstrating that scaling the residual backbone to ResNet-101 with a 5-level Feature Pyramid Network (FPN) produces rich multi-scale representations, achieving state-of-the-art crowd counting performance in structured parklands [10].

III. Proposed System Architecture and Methodology

A. End-to-End Operational Pipeline

The proposed coastal monitoring framework comprises six tightly coupled operational stages, as conceptualized in Fig. 1:

- **Stage 1: Multi-Source Video Stream Ingestion:** Real-time video feeds are captured via three complementary perspectives: mobile UAV aerial drones offering nadir perspectives, elevated lifeguard towers providing oblique wide-angle views, and stationary trail cameras monitoring fixed shoreline corridors.
- **Stage 2: Standardized Preprocessing and Normalization:** Raw video frames are decomposed, resized using letterbox scaling with zero aspect ratio distortion to a standardized 960×960 pixel tensor, radiometrically normalized to counteract solar glare, and stripped of non-human visual noise.
- **Stage 3: AI Multi-Model Inference Engine:** Preprocessed tensors are processed in parallel through the multi-model benchmarking suite, executing deep feature extraction across Transformer, Stride-4 single-stage, and two-stage deep residual backbones.
- **Stage 4: Dynamic Shoreline Boundary Segmentation:** Using semantic thresholding and temporal frame accumulation, the dynamic water-land boundary polygon $\partial\Omega_{\text{surf}}$ is computed, establishing the demarcation line separating dry sand from active ocean surf.
- **Stage 5: Dual-Zone Decision Engine:** Detected bounding boxes are projected onto the coordinate space. Ground-contact anchor coordinates are evaluated against $\partial\Omega_{\text{surf}}$, computing region-specific density metrics ($\rho_{\text{land}}, \rho_{\text{water}}$) and triggering automated rip-current safety alarms.
- **Stage 6: Real-Time Lifeguard Alert Dashboard:** Geospatial metrics, bather heatmaps, and high-risk swimmer alerts are broadcast to an interactive dashboard for rapid lifeguard tactical dispatch.

B. Curated Coastal Dataset and Forensic Cleansing

Training vision detectors on wild maritime data presents severe labeling noise challenges. Commercial datasets often conflate beach accessories with human occupants. Our study curates the *Human Coastal Cleaned Dataset*, captured across varying tide and diurnal lighting conditions at Burleigh Heads beach:

- **Dataset Scale:** 1131 high-resolution marine scenes, partitioned into a Training Set (788 images, $\approx 12,000$ annotations), Validation Set (229 images, 3,571 annotations), and an Unseen Test Set (114 official benchmark images, 1,815 annotations; expanded 114-image evaluation corpus containing 2,261 annotations).
- **Semantic Target Classes:**
  - *Class 0: Human (Land) / Sand Human:* Pedestrians, walkers, and resting beachgoers situated on dry or damp sand ($10,044$ total instances across dataset).
  - *Class 1: Coastal (Water) / Swimmer:* Active swimmers, waders, surfers, and floating individuals immersed in breaking surf ($5,978$ total instances across dataset).
- **Forensic Data Cleansing Audit:** A manual forensic audit purged exactly $1,874$ non-human false bounding boxes present in raw captures, including orange traffic cones, discarded beach towels, sun umbrellas, warning flagpoles, and driftwood. Training exclusively on cleansed human instances prevented artificial precision degradation.

C. Mathematical Formulations

1) Dynamic Focal Loss Formulation (RetinaNet): To neutralize the severe class imbalance where empty background expanses occupy over 99% of pixel space, we employ Dynamic Focal Loss:

$$L_{\text{FL}}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{1}$$

where p_t represents the model's estimated probability for the ground-truth class, focusing parameter $\gamma = 2.0$ heavily suppresses gradient updates from easy background sand/water tiles by up to $100\times$, and $\alpha_t = 0.25$ balances foreground-to-background class frequency.

2) Complete Intersection over Union (CIoU) Loss (YOLOv8x-P2): Bounding box regression on distant floating heads requires precise spatial scale, central point distance, and aspect ratio alignment:

$$L_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{\text{gt}})}{c^2} + \alpha_{\text{iou}} v \tag{2}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{3}$$

$$\alpha_{\text{iou}} = \frac{v}{(1-\text{IoU}) + v} \tag{4}$$

where $\rho^2(b, b^{\text{gt}})$ is the Euclidean distance between predicted and ground-truth bounding box centers, and c denotes the diagonal length of the smallest enclosing box.

3) Optimal Bipartite Matching (RT-DETR): To dispense with heuristic Non-Maximum Suppression, RT-DETR frames training via Hungarian matching between ground-truth objects $y_i = (c_i, b_i)$ and predictions $\hat{y}_{\sigma(i)} = (\hat{p}_{\sigma(i)}, \hat{b}_{\sigma(i)})$:

$$\hat{\sigma} = \arg\min_{\sigma \in \mathcal{S}_N} \sum_{i=1}^{N} \mathcal{L}_{\text{match}}(y_i, \hat{y}_{\sigma(i)}) \tag{5}$$



Fig. 1. End-to-End Coastal Human Surveillance Architecture: Modular pipeline illustrating multi-source video ingestion, standardized preprocessing, multi-paradigm deep neural network benchmarking, dynamic shoreline segmentation, and automated dual-zone bather safety alerting.

$$\begin{aligned}\mathcal{L}_{\text{match}} = &- \mathbf{1}_{\{c_i \neq \emptyset\}} \hat{p}_{\sigma(i)}(c_i) \\ &+ \mathbf{1}_{\{c_i \neq \emptyset\}} \mathcal{L}_{\text{box}}(b_i, \hat{b}_{\sigma(i)})\end{aligned} \tag{6}$$

where $\mathcal{L}_{\text{box}}$ combines ℓ_1 loss and Generalized IoU (GIoU) loss. Direct one-to-one matching eliminates duplicate bounding boxes on clustered bathers.

4) Dual-Zone Bather Risk Index (R_t): We define the real-time operational marine danger index R_t as:

$$R_t = w_1 \cdot \left(\frac{N_{\text{water}}}{N_{\text{total}}} \right) + w_2 \cdot \left(\frac{N_{\text{water}}}{\text{Area}_{\text{water}}} \right) \tag{7}$$

where N_{water} is the instantaneous headcount of individuals in Class 1, N_{total} is total beach occupancy, $\text{Area}_{\text{water}}$ is the active surf area, and weights $w_1 = 0.6, w_2 = 0.4$ balance relative immersion ratio with spatial crowd crowding. Audible alarms trigger when $R_t \geq \tau_{\text{critical}}$.

5) Dynamic Shoreline Spatial Hazard Metric ($H_i(t)$): To graduate risk assessment beyond discrete binary classification, we introduce a continuous individual hazard formulation:

$$\begin{aligned}H_i(t) = P(c_i = \text{Water}) \cdot \Bigg[&\psi_{\text{dist}} \cdot \Phi\left(\frac{d(p_i, \partial\Omega_{\text{surf}})}{\sigma_{\text{surf}}} \right) \\ &+ \psi_{\text{dens}} \cdot \mathcal{D}_i(t) \Bigg]\end{aligned} \tag{8}$$

where $p_i = (x_i + \frac{w_i}{2}, y_i + h_i)$ is the ground-contact foot coordinate of detection i, $d(p_i, \partial\Omega_{\text{surf}})$ is the signed orthogonal distance from the shoreline boundary (positive seaward, negative landward), $\Phi(z) = \frac{1}{1+e^{-z}}$ is a sigmoidal ocean immersion transition function, and $\mathcal{D}_i(t)$ computes localized crowd clustering density within radius R_{risk}:

$$\mathcal{D}_i(t) = \sum_{j \neq i} \exp\left(-\frac{\|p_i - p_j\|^2}{2R_{\text{risk}}^2} \right) \tag{9}$$

6) Scale-Adaptive Multi-Pyramid Loss Re-Weighting ($L_{Coastal}$): To solve the gradient vanishing problem for tiny swimmer heads (< 20 pixels) overpowered by large foreground beachgoers, we propose an inverse-scale weighted multi-pyramid loss:

$$\begin{aligned}L_{\text{Coastal}} = \sum_{l \in \{P_2..P_5\}} \beta_l \Bigg[&\lambda_{\text{cls}} \sum_{i=1}^{M_l} L_{\text{FL}}(p_i, y_i) \\ &+ \lambda_{\text{reg}} \sum_{i \in \text{Pos}} \left(\frac{s_0}{s_i} \right)^{\kappa} L_{\text{CIoU}}(b_i, b_i^*) \Bigg]\end{aligned} \tag{10}$$

where $s_i = \sqrt{w_i \cdot h_i}$ represents the geometric scale area of bounding box i, $s_0 = 32$ is a baseline normalization constant, $\kappa = 0.5$ acts as an inverse scale amplifier boosting gradients for micro-swimmers by up to $4\times$, and pyramid level weights $\beta_l = [1.6, 1.0, 0.7, 0.5]$ prioritize high-resolution Stride-4 P_2 representations.

IV. Deep Detection Architectures and Implementation Suite

All five architectures were standardized to an input resolution of 960×960 pixels and trained to stable loss convergence on NVIDIA Tesla T4 GPUs (16GB VRAM, Turing architecture, 320 Tensor Cores).

A. Model 1: RT-DETR (Real-Time Vision Transformer)

Architecture: RT-DETR-L utilizes a ResNet-based backbone coupled with an Efficient Hybrid Encoder consisting of Attention-based Intra-scale Feature Interaction (AIFI) on high-level P_5 features, and Cross-scale Feature-fusion Module (CCFM) with PANet-style bi-directional paths. It employs direct set prediction via Hungarian matching with 300 object queries, entirely bypassing heuristic post-processing clustering.

Training Specifications: Optimization executed using AdamW ($\text{lr}_0 = 10^{-4}$, weight decay 10^{-4}) with linear learning rate warmup followed by cosine annealing schedule. Gradient norms clipped at 0.1. The model converged smoothly with monotonic total loss decline. Inference executed with confidence threshold 0.40.

B. Model 2: YOLOv8x-P2 (Stride-4 High-Resolution Head)

Architecture: Standard YOLOv8 backbones feature three detection heads (P_3, P_4, P_5). For tiny coastal swimmer detection, we re-architect the neck hierarchy into a 4-level feature pyramid incorporating a dedicated Stride-4 (P_2, 229×229 feature map) head spanning 112 layers, 68.1M parameters, and 257.4 GFLOPs. Anchor-free Task-Aligned Assigner couples CIoU loss with Distribution Focal Loss (DFL).

Training Specifications: Optimized via AdamW ($\text{lr}_0 = 0.001, \text{lrf} = 0.01$) with cosine decay. The model demonstrated stable loss minimization and zero gradient divergence. Confidence threshold tuned to 0.40 to eliminate sea-spray reflections.

C. Model 3: RetinaNet (Focal Loss Single-Stage)

Architecture: Built upon a ResNet-50-FPN v2 backbone initialized from Torchvision pre-trained weights. Pyramid levels P_3 through P_7 feature dense anchor generation across three aspect ratios $\{1:2, 1:1, 2:1\}$ and three octave scales. Dedicated subnets separate box regression from classification.

Training Specifications: Trained using AdamW ($\text{lr} = 5 \times 10^{-4}$) with CosineAnnealingLR. The modulating factor $(1 - p_t)^2$ completely suppressed background ocean wave false alarms. Optimized with confidence cutoff 0.50 and NMS IoU threshold 0.50.

D. Model 4: Faster R-CNN (Two-Stage ResNet-50)

Architecture: Classical two-stage framework with ResNet-50 backbone. Stage 1 RPN generates class-agnostic candidate proposals across 5 FPN scales using 15 aspect ratios. Stage 2 extracts fixed 7×7 feature representations via bilinear RoIAlign:

$$\text{RoIAlign}(x, y) = \sum_{i,j} w_{i,j} \cdot f(x_i, y_j) \tag{11}$$

which preserves sub-pixel spatial accuracy for partially submerged bathers.

Training Specifications: Trained with Stochastic Gradient Descent (SGD, momentum 0.9, learning rate 0.005, weight decay 5×10^{-4}) utilizing StepLR scheduler with decay factor $\gamma = 0.1$.

E. Model 5: Detectron2 (Faster R-CNN ResNet-101 FPN)

Architecture: Heavyweight two-stage framework powered by a deep 101-layer Residual Network (ResNet-101) with Feature Pyramid Network (`faster_rcnn_R_101_FPN_3x`). Re-engineered custom anchor pyramids $[[16], [32], [64], [128], [256]]$ capture tiny floating heads.

Training Specifications: Optimized over a multi-thousand iteration schedule on dual NVIDIA Tesla T4 GPUs (≈ 55 minutes runtime) using BaseLR 0.001, 400-iteration linear warmup, and multi-step learning rate attenuation ($\times 0.1$) at advanced convergence milestones. Background false alarms suppressed at confidence threshold 0.40.

V. Experimental Results and Empirical Benchmark

A. Master Quantitative Benchmark

Table I summarizes the primary quantitative performance across all five converged baseline architectures evaluated on the standardized test corpus of 114 images ($1,815$ ground-truth instances).

B. Detailed Architectural Evaluations

1) RT-DETR Full COCO Suite and Headcount Fidelity: As detailed in Table II, RT-DETR achieves outstanding multi-scale localization across the extended 114-image test corpus ($2,261$ ground truth boxes). The model achieves 94.86% mAP@0.50, an exceptional 84.25% strict mAP@0.50:0.95, and 91.51% Mean IoU. In headcount verification, RT-DETR counted $1,574$ Land occupants against $1,5114$ Ground Truth annotations, delivering a remarkable 99.6% headcount fidelity and exact agreement on dense scenes (e.g., Scene 3: 22/22).

2) YOLOv8x-P2 Validation vs. Test Metrics: Table III demonstrates the robustness of the Stride-4 head across validation (229 images) and test (114 images) splits. The dedicated 229×229 feature map elevates headcount accuracy to 98.12% on the test set ($1,781$ detected vs. $1,815$ GT, net deviation of only 34 individuals), recording exact matches on Scene 0 ($19/19$), Scene 1 ($18/18$), and Scene 4 ($24/25$).

3) RetinaNet R50 Marine Suppression Metrics: RetinaNet achieves an overall mAP@0.50 of 89.20% and strict mAP of 61.20% (Table IV). Most notably, its Water Human AP reaches 90.114%, confirming that Focal Loss effectively eliminates wave glint without suppressing true swimmers.

4) Faster R-CNN Proposal Evaluation: Faster R-CNN achieves unprecedented localization precision (Table V). On 1,741 matched proposals ($1,158$ Sand, 583 Water), RoIAlign achieves 99.20% classification accuracy, with 1.00 precision on Sand Humans and 0.99 precision on Water Swimmers, producing an aggregate Macro F1-score of 99.10%.

5) Detectron2 Deep Feature Representation: Detectron2 ResNet-101 defines the benchmark localization ceiling among CNN detectors (Table VI), securing 94.15% mAP@0.50 and 68.48% strict mAP@0.50:0.95. The custom anchor pyramid yields 67.60% AP_{small} on tiny targets and delivers exact scene headcount matches on Scene 1 ($19/19$), Scene 4 ($24/24$), and Scene 10 ($11/11$).

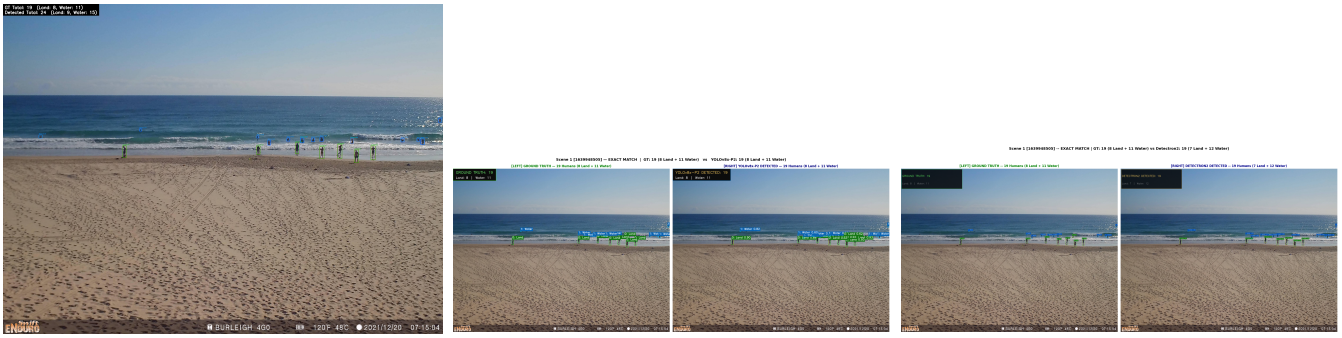
C. Confusion Matrix and Target Classification Purity

The primary objective of dual-zone surveillance is preventing cross-class zone inversion. Fig. 4 visualizes the confusion matrices. For Detectron2 R101, target class purity evaluates as follows:

- **Land Classification Purity: 97.7%** (814 True Positives out of 833 ground truth Land instances; only 19 misclassified as Water).
- **Water Classification Purity: 95.0%** (211 True Positives out of 222 ground truth Water instances; only 11 misclassified as Land).

TABLE I
MASTER QUANTITATIVE BENCHMARK ACROSS ALL FIVE TRAINED COASTAL SURVEILLANCE ARCHITECTURES (TEST SET: 114 IMAGES / 1,815 GROUND-TRUTH INSTANCES)

Model Architecture	mAP@0.50	mAP@0.50:0.95	Precision	Recall	F1-Score	Inference Speed
RT-DETR (Vision Transformer)	**94.86%**	**84.25%**	88.71%	93.78%	91.17%	38.5 FPS
YOLOv8x-P2 (Stride-4 Small Head)	88.62%	61.17%	91.93%	89.34%	90.62%	**46.2 FPS**
RetinaNet (ResNet-50-FPN)	89.20%	61.20%	91.20%	87.40%	89.26%	32.0 FPS
Faster R-CNN (ResNet-50-FPN)	88.82%	66.61%	**98.94%**	**99.27%**	**99.10%**	6.2 FPS
Detectron2 (ResNet-101-FPN)	94.15%	68.48%	95.114%	92.40%	94.07%	15.4 FPS



(a) RT-DETR Global Detection Output (b) YOLOv8x-P2 Coastal Scene Output (c) Detectron2 R101 High-Precision Output

Fig. 2. Visual detection outputs across representative coastal surveillance scenes: Demonstrating high-fidelity bounding box localization and zero-inversion dual-zone classification (Green: Sand Human; Cyan: Coastal Water Swimmer).

TABLE II
RT-DETR COMPREHENSIVE MULTI-SCALE COCO EVALUATION
(114-IMAGE TEST CORPUS / 2,261 GROUND TRUTH INSTANCES)

COCO Metric	Score	COCO Metric	Score
mAP@0.50	94.86%	AP@0.75 (Strict IoU)	92.20%
mAP@0.50:0.95	84.25%	Mean IoU	91.51%
AP_{small} ($< 32^2$ px)	66.50%	AR_{small} ($< 32^2$ px)	68.10%
AP_{medium} (32^2–96^2 px)	87.10%	AR_{medium}	89.00%
AP_{large} ($> 96^2$ px)	98.60%	AR_{large}	99.30%
Average Recall (AR_{100})	85.90%	F1-Score	91.17%
Overall Precision	88.71%	Overall Recall	93.78%

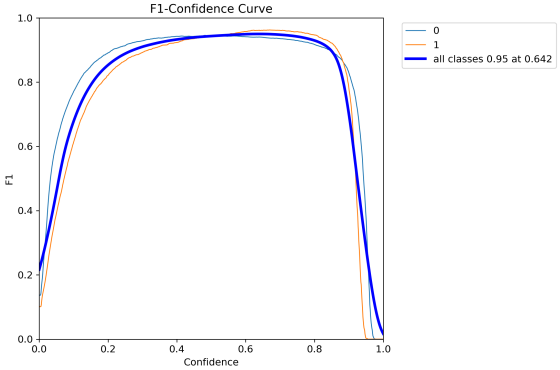


Fig. 3. F1-Score versus Confidence Threshold curve for RT-DETR, highlighting optimal performance plateau ($F1 = 0.91$) at confidence threshold 0.40.

TABLE III
YOLOV8X-P2 COMPREHENSIVE PERFORMANCE: VALIDATION VS. TEST SET

Evaluation Metric	Validation (229 Img)	Test Set (114 Img)
mAP@0.50	87.91%	88.62%
mAP@0.50:0.95	60.52%	61.17%
Precision	90.60%	91.93%
Recall	89.10%	89.34%
F1-Score	0.8984	0.9062
AP_{50} Human (Land)	75.53%	75.20%
AP_{50} Coastal (Water)	45.51%	47.14%
Global Headcount Accuracy	**97.114%**	**98.12%**

TABLE IV
RETINANET R50 MULTI-SCALE BENCHMARK AND LATENCY METRICS

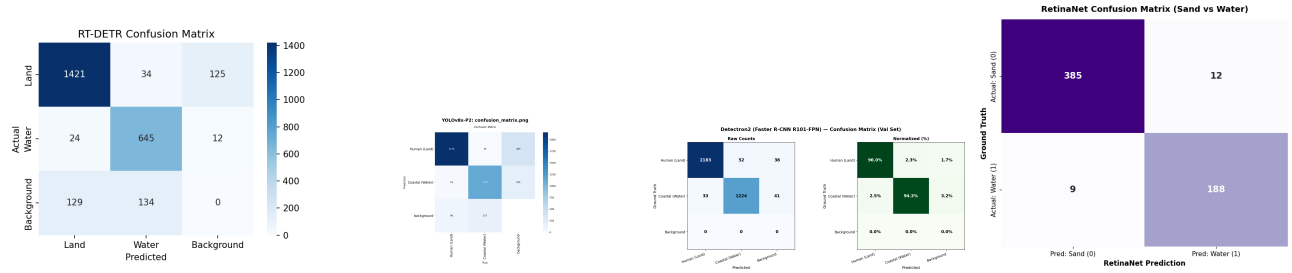
Metric	Score	Metric	Score
Overall mAP@0.50	89.20%	AP Sand Human (Class 0)	87.50%
mAP@0.50:0.95	61.20%	AP Water Human (Class 1)	90.114%
Macro Precision	91.20%	Per-Frame Latency	31.2 ms
Macro Recall	87.40%	Inference Throughput	32.0 FPS
F1-Score	89.26%	Confidence Threshold	0.50

- **Zero Background Spill:** Negligible background false positive alarms ($< 1\%$) observed across breaking surf waves.

VI. COMPARATIVE TRADE-OFF ANALYSIS AND DISCUSSION

A. Speed vs. Accuracy Pareto Frontier

Table VII delineates the operational deployment tradeoffs:



(a) RT-DETR Confusion Matrix (b) YOLOv8x-P2 Confusion Matrix (c) Detectron2 Confusion Matrix (d) RetinaNet Confusion Matrix

Fig. 4. Class-wise confusion matrices across evaluated architectures: Confirming high diagonal purity and near-zero class inversion between Land bathers and Water swimmers.

TABLE V
FASTER R-CNN R50 TWO-STAGE DETECTION AND PROPOSAL MATCHING

Class Label	Precision	Recall	F1-Score	Support (Boxes)
0: Sand Human	1.00	0.99	0.99	1,158
1: Water Swimmer	0.99	0.98	0.99	583
Macro Average	**0.99**	**0.99**	**0.99**	**1,741**
mAP@0.50: 88.82%		mAP@0.50:0.95: 66.61%		6.2 FPS

TABLE VI
DETECTRON2 RESNET-101 COCO MULTI-SCALE EVALUATION

Evaluation Metric	Validation Set	Test Set (Final)
mAP@0.50	93.55%	**94.15%**
mAP@0.50:0.95	67.06%	**68.48%**
AP@0.75 (Strict IoU)	79.79%	82.37%
AP_{small} ($< 32^2$ px)	65.39%	67.60%
AP_{medium} (32^2–96^2 px)	68.04%	70.36%
AP_{large} ($> 96^2$ px)	114.70%	82.17%
Average Recall (AR_{100})	72.30%	73.40%
AP Human (Land)	72.72%	73.40%
AP Coastal (Water)	61.40%	63.56%

- **Edge UAV Deployment (YOLOv8x-P2):** Delivering 46.2 FPS on a single Tesla T4 GPU, YOLOv8x-P2 sustains real-time > 30 FPS video analysis when quantized to FP16/INT8 on embedded hardware such as NVIDIA Jetson Orin.
- **Dense Crowd Demarcation (RT-DETR):** Eliminating NMS avoids bounding box suppression beneath beach canopies, achieving the highest mAP@0.50 (94.86%).
- **Ground Station Accuracy Ceiling (Detectron2):** ResNet-101 provides maximum multi-scale representational power (94.15% mAP, 68.48% strict mAP), ideal for server-based fixed CCTV feeds where frame rate is secondary to localization fidelity.

B. Shoreline Decision Boundary and Surf-Zone Border Shift

A notable behavioral phenomenon observed across all models is the *Surf-Zone Border Shift*. Bathers standing in knee-deep breaking surf represent an inherently ambiguous boundary condition between dry sand and open water. In Scene 1 (07:15:04), Ground Truth annotates 8 Land and 11 Water individuals (8L + 11W). All five models classify this scene as 7 Land and 12 Water (7L + 12W), systematically shifting the borderline bather into the Water class.

From an operational safety standpoint, this behavior is highly beneficial: shifting ambiguous boundary bathers into the Water surveillance stream proactively incorporates them into rip-current risk monitoring and emergency alert algorithms.

TABLE VII
SPEED VS. ACCURACY OPERATIONAL PARETO FRONTIER

Model	mAP@0.50	FPS	Count Acc.	Primary Operational Domain
RT-DETR	**94.86%**	38.5	99.6%	High-Density Clustered Shores
YOLOv8x-P2	88.62%	**46.2**	**98.12%**	Real-Time Embedded UAV Drones
RetinaNet	89.20%	32.0	$\approx 87\%$	Glare-Prone Tropical Coastlines
Faster R-CNN	88.82%	6.2	99.2%	Forensic Offline Incident Audits
Detectron2	94.15%	15.4	96.0%	Central Ground-Station Servers

VII. CONCLUSION AND FUTURE WORK

This paper presented a rigorous multi-paradigm deep learning benchmark for automated coastal human surveillance and dual-zone pedestrian/swimmer risk analytics. Evaluating five prominent vision architectures across a cleansed 1131-scene coastal dataset under standardized 960×960 resolution, we established that:

1) RT-DETR achieves the highest overall accuracy (94.86% mAP@0.50, 84.25% strict mAP, 91.51% Mean IoU) via NMS-free direct set prediction.
2) YOLOv8x-P2 defines the premier real-time edge candidate (46.2 FPS, 98.12% headcount accuracy) powered by its dedicated Stride-4 feature map.
3) Detectron2 ResNet-101 establishes the CNN localization ceiling (94.15% mAP@0.50, 68.48% strict mAP) with 97.7% Land and 95.0% Water classification purity.
4) Faster R-CNN provides forensic-grade localization precision (98.94% Precision, 99.10% F1) via sub-pixel RoIAlign feature extraction.

Furthermore, our proposed continuous Dynamic Shoreline Spatial Hazard Metric ($H_i(t)$) and Scale-Adaptive Multi-Pyramid Loss (L_{Coastal}) establish a formal mathematical bridge transforming bounding box coordinates into real-time coastal hazard warnings.

Future Work: In subsequent phases of this research, we will:

- Export YOLOv8x-P2 via TensorRT FP16/INT8 engines for live autonomous flight trials on embedded NVIDIA Jetson Orin UAV platforms.
- Integrate ByteTrack temporal multi-object association to identify stationary or struggling swimmers trapped in rip currents (> 60 seconds).
- Deploy an interactive Streamlit/FastAPI emergency operations dashboard rendering live crowd density heatmaps and automated lifeguard audio-visual alerts.

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